AAL APPAREL

Q4 2020 Sales Analysis Report

Prepared for: Head of Sales & Marketing (S&M)

Scope: October – December 2020 | All Australian States | All Demographic Groups

Total Q4 Revenue	Total Units Sold	Total Transactions	States Covered
$340.3M	**136,121**	**7,560**	**7**
Oct–Dec 2020	*All states & groups*	*Avg $45,014 each*	*WA NT SA VIC QLD NSW TAS*

Dataset: AusApparalSales4thQrt2020.csv | Notebook: AAL_Sales_Analysis.ipynb

1. Executive Summary

AAL Apparel generated $340.3 million in total revenue across Australia during Q4 2020 (October–December), spanning 7,560 recorded transactions across seven states and four demographic groups. This report synthesises all analytical findings from the data wrangling, statistical analysis, and visualisation phases and translates them into actionable recommendations for the Sales & Marketing leadership team.

Three headline insights emerge:

- Geographic concentration is extreme — Victoria alone accounts for 31% of all Q4 revenue ($105.6M), while the four lowest-performing states together contribute less than VIC's total.
- Demographic demand is remarkably balanced — Men, Women, Kids, and Seniors each contribute approximately 25% of revenue, confirming AAL's universal brand appeal.
- Morning is the technical peak sales window, yet the difference across all three time slots is less than 2%, signalling untapped potential in evening and off-peak periods.

The December surge ($135.3M vs November's $90.7M) signals strong festive-season demand and is the single most actionable planning anchor for 2021 investment.

2. Dataset Overview & Data Wrangling Summary

2.1 Dataset Structure

The dataset contains 7,560 rows and 6 columns covering daily apparel sales across Australian states for Q4 2020:

Column	Data Type	Description
Date	datetime64	Transaction date — October 1 to December 31, 2020
Time	object	Time of day: Morning, Afternoon, Evening
State	object	Australian state: WA, NT, SA, VIC, QLD, NSW, TAS
Group	object	Demographic group: Kids, Men, Women, Seniors
Unit	int64	Number of units sold per transaction
Sales	int64	Revenue in AUD per transaction

2.2 Data Quality Findings

- **No missing values**: All 7,560 records complete across all 6 columns (confirmed via df.isna().sum()).
- **Date type corrected**: The Date column was stored as an object (string) and converted to datetime64 to enable time-series resampling and chronological analysis.
- **Whitespace errors detected and resolved:** All three categorical columns (Time, State, Group) contained leading spaces (e.g., ' Morning'). These were stripped using .str.strip(), restoring 7 clean state codes, 3 time slots, and 4 group labels.
- **Outliers retained**: Box plots on Sales and Unit revealed high-value outliers (Sales up to $162,500; Units up to 65). These are valid high-capacity bulk transactions — not data errors — and were retained.

2.3 Normalization

Min-Max Normalization (sklearn MinMaxScaler) was applied to Sales and Unit columns, scaling both to a [0,1] range. This was necessary because Sales values range in the tens of thousands

while Units are in the double digits — a difference that would distort any cross-variable comparison. Original values were preserved in a separate copy (df_norm) for all business reporting.

3. Descriptive Statistical Analysis

A Shapiro-Wilk normality test confirmed that neither Sales nor Units follow a normal distribution ($W = 0.896$, $p < 0.05$ for both). This has a direct and important implication: the mean is not a reliable measure of a 'typical' transaction.

Statistic	Sales (AUD)	Units Sold	Business Interpretation
Mean	**$45,014**	18.0 units	Inflated by high-value outliers — do not use for targets
Median	**$35,000**	14.0 units	Best benchmark — representative of the typical transaction
Mode	$22,500	9 units	Most common transaction size — the 'everyday' purchase
Std Deviation	$32,254	12.9 units	High variability — broad spread of transaction sizes
Min	$5,000	2 units	Smallest recorded transaction
Max	$162,500	65 units	Largest recorded transaction
Skewness	1.09	1.09	Right-skewed — mean pulled upward by high-value deals
Kurtosis	0.66	0.66	Platykurtic — broad, flat distribution; revenue is diverse

Key implication: The mean overstates a typical transaction by nearly $10,000. All internal sales targets should be benchmarked against the median ($35,000), not the mean. Using the mean will create systematically unrealistic expectations for the majority of stores and sales staff.

The near-identical skewness (1.09) and kurtosis (0.66) for both Sales and Units confirms the constant price-per-unit finding from the regression analysis — revenue and volume are effectively the same dimension in this dataset.

4. Geographic Analysis — State-by-State Performance

State-level analysis reveals a highly uneven revenue distribution.

#	State	Total Sales (AUD)	Units Sold	Q4 Share	Tier
1	VIC — Victoria	**$105,565,000**	42,226	**31.0%**	Top
2	NSW — New South Wales	$74,970,000	29,988	22.0%	High
3	SA — South Australia	$58,857,500	23,543	17.3%	Mid
4	QLD — Queensland	$33,417,500	13,367	9.8%	Low
5	TAS — Tasmania	$22,760,000	9,104	6.7%	Low
6	NT — Northern Territory	$22,580,000	9,032	6.6%	Low
7	WA — Western Australia	**$22,152,500**	8,861	**6.5%**	Lowest

- VIC generates $105.6M — nearly 4.8x the revenue of WA ($22.2M). This is not a marginal gap; it is a structural market penetration difference.
- Top 3 states (VIC, NSW, SA) together generate $239.4M — 70.3% of all Q4 revenue.
- Bottom 4 states (QLD, TAS, NT, WA) generate only $100.9M combined — 29.7% — despite collectively covering the majority of Australia's land mass.
- QLD at $33.4M is the most notable underperformer relative to population size. It represents the highest-potential expansion market.
- The group-by-state heatmap in the notebook confirms that VIC leads in every single demographic group, and WA trails in all. The geographic gap is systemic — not driven by any one product category.

5. Demographic Group Analysis

AAL caters to four demographic segments: Kids, Men, Women, and Seniors. The Q4 data reveals the brand's most striking characteristic: near-perfect revenue balance across all four groups.

Men	Women	Kids	Seniors
$85.75M	**$85.45M**	**$85.07M**	**$84.04M**
34,300 units — Highest	*~25.1% share*	*~25.0% share*	*33,615 units — Lowest*

#	Group	Total Sales (AUD)	Units Sold	Revenue Share	vs. Leader	Gap ($)
1	Men	$85,750,000	34,300	25.2%	—	—
2	Women	$85,447,500	34,179	25.1%	-0.4%	-$302,500
3	Kids	$85,067,500	34,027	25.0%	-0.8%	-$682,500
4	Seniors	**$84,037,500**	33,615	**24.7%**	-2.0%	-$1,712,500

- The spread between highest (Men, $85.75M) and lowest (Seniors, $84.04M) is only $1.71M — less than 2% of the top group. This is an exceptionally tight distribution.
- Universal brand appeal confirmed: AAL's product range successfully penetrates all four demographic segments with near-equal effectiveness. This is a genuine competitive strength.
- This balance also means there is no dominant segment to 'double down' on for quick revenue gains. Growth must come from geographic expansion and transaction frequency, not segment mix shifts.
- Seniors trail by the widest margin. This may reflect product range gaps, pricing sensitivity in the 65+ bracket, or lower digital engagement. A targeted Seniors initiative could close the gap incrementally.

6. Time-of-Day Analysis

The dataset captures sales across three time windows: Morning, Afternoon, and Evening. This dimension is critical for designing hyper-personalisation campaigns and Next Best Offer (NBO) programs for the S&M team.

Morning	Afternoon	Evening
$114.2M	**$114.0M**	**$112.1M**
45,683 units — Peak	*45,603 units*	*44,835 units — Off-Peak*

Time Slot	Total Sales	Units Sold	Q4 Share
Morning	$114,207,500	45,683	33.6%
Afternoon	$114,007,500	45,603	33.5%
Evening	**$112,087,500**	44,835	32.9%

- Morning leads technically ($114.2M), but the gap over Afternoon is only $200,000 — a 0.2% difference. All three windows are nearly statistically identical at the aggregate level.
- Evening is the only slot below the daily average (index 98.7). This is the primary time-of-day opportunity.
- The flat time distribution is surprising. Most retail sectors show pronounced peak windows. AAL's near-uniform spread suggests that customer buying is not time-constrained — which is an opportunity for targeted promotions to create deliberate peaks.
- Strategic recommendation: Deploy 'After Hours' push notifications at 5–7pm with time-limited offers. Even a 3% Evening revenue lift translates to ~$3.4M additional Q4 revenue at current scale.

7. Time-Series Analysis: Weekly & Monthly Trends

Q4 Total Revenue	Best Month	Weakest Month	Peak Week
$340.3M	**December**	**November**	**Dec 27+**
Oct–Dec 2020	*$135.3M — festive peak*	*$90.7M — seasonal trough*	*$37.5M+ Christmas surge*

7.1 Monthly Breakdown

Month	Total Sales (AUD)	Units Sold	Q4 Share	vs Prior Month
October 2020	$114,290,000	45,716	33.6%	— (baseline)
November 2020	**$90,682,500**	36,273	26.6%	-20.7%
December 2020	**$135,330,000**	54,132	39.8%	+49.2%

- October opened at $114.3M — a solid baseline reflecting end-of-season clearance and early summer apparel demand across Australia.
- November dipped sharply to $90.7M (-20.7%) — the post-October spending lull, representing a $23.6M gap versus October. This is the most significant recoverable opportunity in the quarterly cycle.
- December surged to $135.3M (+49.2% vs November) — the strongest single month of Q4, driven by Christmas and end-of-year gifting. This $44.6M swing from November to December is the dataset's most dramatic pattern.

7.2 Weekly Trend

Week Ending	Sales (AUD)	Commentary
Oct 4 (partial)	$15,045,000	Short first week — first 4 trading days of Q4 only
Oct 11	$27,002,500	First full week — strong October baseline established
Oct 18	$26,640,000	Consistent mid-October performance

Week Ending	Sales (AUD)	Commentary
Oct 25	$26,815,000	October closes steady at ~$27M/week
Nov 1–Nov 29	**~$21M/week**	November trough — 5 consecutive subdued weeks averaging $21M
Dec 6	$29,622,500	December recovery begins — +41% vs November average
Dec 13	$31,525,000	Festive ramp-up accelerates
Dec 20+	**$37,500,000+**	Christmas peak — highest weekly revenue of Q4

The five-week November trough (averaging ~$21M/week vs October's ~$27M/week) represents the clearest tactical opportunity. A November Activation Campaign — anchored around Black Friday, Cyber Monday, and pre-Christmas positioning — could recover a meaningful share of this gap.

8. Units–Sales Relationship & Pricing Insight

A regression scatter plot (sns.regplot) examined the relationship between units sold and total sales revenue per transaction. The finding has important strategic implications:

- Near-perfect linear correlation — Data points cluster almost exactly on the regression line, with virtually no scatter or noise ($R^2 \approx 1.0$).
- Constant price-per-unit confirmed — The uniformity indicates that price per unit is consistent across virtually all transactions. There is negligible pricing variation in the dataset.
- Revenue = Volume — Because price is constant, total revenue is almost entirely a function of volume (units sold). Growing revenue means growing units sold.

Strategic Implication	Detail
Pricing Opportunity	Uniform pricing limits revenue upside. Introducing tiered pricing (e.g., premium lines for VIC/NSW, value-entry lines for WA/NT) could improve margins without volume changes.
Primary Growth Lever	Volume is the only driver of revenue at current pricing. State expansion programs must focus on increasing transaction frequency and basket size (units per transaction).
NBO Design Focus	Since all customers pay the same price, Next Best Offer programs should aim to increase units per basket (cross-sell / up-sell to larger bundle sizes) rather than offering price-based discounts.

9. Strategic Recommendations

The following recommendations are presented in priority order, based on estimated revenue impact and implementation feasibility:

Priority 1: November Revenue Recovery Programme

The November trough ($90.7M) is the single largest recoverable revenue gap in Q4. A dedicated campaign should be built around three anchors:

- Black Friday (last Friday of November) — Flash sales across all states with group-targeted promotions (e.g., Kids outerwear, Men's workwear bundles).
- Cyber Week — App and online-exclusive offers to capture digital-first shoppers in the 7–10 days post-Black Friday.
- Pre-Christmas Preview — Exclusive early access to December collections for loyalty members, converting November browsers into December buyers.

Target: Recover 30–50% of the October–November gap, adding $7–12M to next year's Q4 at minimal incremental cost.

Priority 2: State Expansion — QLD and SA

Queensland ($33.4M) and South Australia ($58.9M) are the clearest candidates for targeted investment. Both have sizable populations but underperform relative to VIC and NSW:

- QLD: The highest-potential underperformer. Launch a city-by-city rollout starting in Brisbane and Gold Coast, focusing on Kids and Men segments.
- SA: Adelaide's mid-tier performance ($58.9M) suggests existing traction. Invest in store experience upgrades and localised brand campaigns to push SA toward the NSW tier.

Avoid spreading investment across all seven states simultaneously. Concentrate on QLD and SA in H1 2021 before reviewing NT and WA.

Priority 3: Evening Activation ('After Hours' Campaign)

Evening is the only time slot underperforming the daily average (index 98.7). Design a targeted Evening push:

- Push notifications at 5–7pm featuring time-limited offers (valid for 2–3 hours only) to create urgency.
- Segment targeting: Women and Seniors — most likely to shop during relaxed evening windows.

- Potential add: co-branded Evening promotions with streaming or delivery platforms popular in the 6–9pm window.

A 3% Evening revenue lift translates to ~$3.4M additional Q4 revenue at current scale.

Priority 4: Seniors Segment Initiative

Seniors trail all other groups by $1.7M (2.0% below Men). While the gap is small, it is the only consistent underperformer:

- Introduce a Seniors Loyalty Programme offering exclusive mid-week access (Tuesday/Wednesday) when foot traffic is lower — creating a premium, unhurried experience.
- Review product range adequacy for 65+ in sizing, comfort, and styling across all key states.
- Pilot in WA and NT first — both states are low overall, and a Seniors-focused drive could serve as a market-entry wedge for broader regional growth.

Priority 5: Reset Internal Targets to Median, Not Mean

All internal sales targets, benchmarks, and performance reviews must be recalibrated to use the median transaction value ($35,000) rather than the mean ($45,014). The right-skewed distribution means the mean is inflated by a small number of bulk transactions. Using the mean creates systematically unrealistic targets for the majority of stores and front-line sales staff.

10. Conclusion

AAL Apparel's Q4 2020 performance demonstrates both the company's core strengths and its clearest growth opportunities. The $340.3M quarterly revenue is underpinned by a remarkably well-balanced demographic base — each of the four customer segments contributing approximately 25% — which represents a genuine competitive advantage and provides resilience against single-segment demand shifts.

However, the geographic concentration (VIC + NSW = 53% of all revenue) and the November seasonal dip ($90.7M — the weakest month by far) represent structural patterns that targeted investment can directly address. The data consistently points to three scalable actions: a dedicated November activation campaign, a focused state expansion into QLD and SA, and an Evening time-slot conversion programme.

The near-perfect linear relationship between units sold and revenue — confirmed by the regression analysis — means that volume growth is the primary and most reliable lever available to the S&M team. Pricing diversification and tiered product lines could unlock a secondary margin lever, but volume remains the foundation.

This analysis was conducted in Python (Pandas, NumPy, SciPy, Seaborn, Matplotlib, Plotly) within a Google Collab Notebook environment, consistent with the Applied Data Science with Python project requirements.

— End of Report —

AAL Apparel | Q4 2020 Sales Analysis | Applied Data Science with Python